

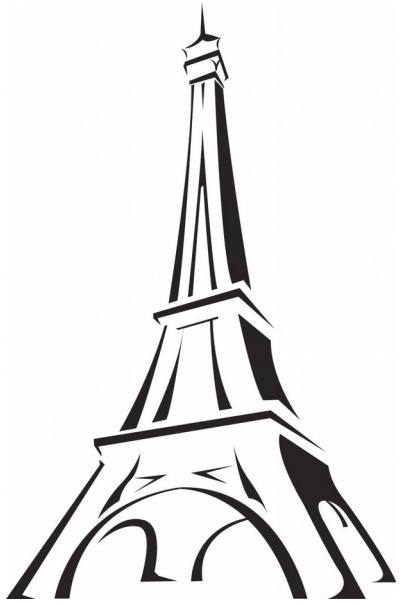


Scientometrics for Success and Influence in the Microsoft Academic Graph

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A summer night in Paris...



My Supervisor is very good! He has 40 hindex.

Wow. Mine has just 25... :/

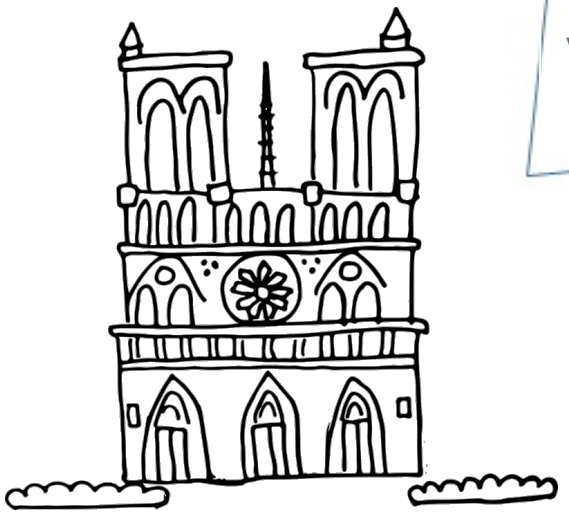
Mine has 50!

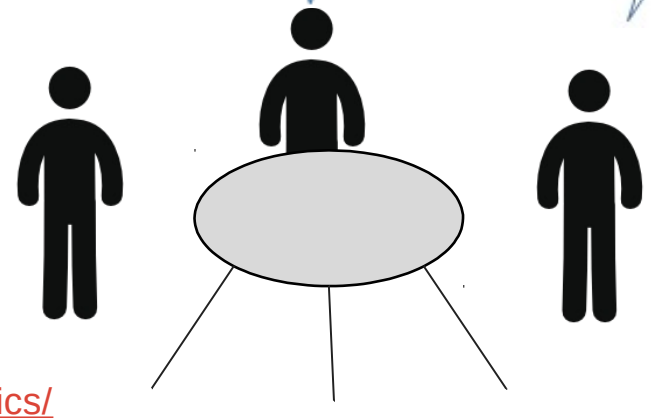
Which field are you in?

But wait! Mine is Math and she is very well-known.

Biology

Oh mine is CS





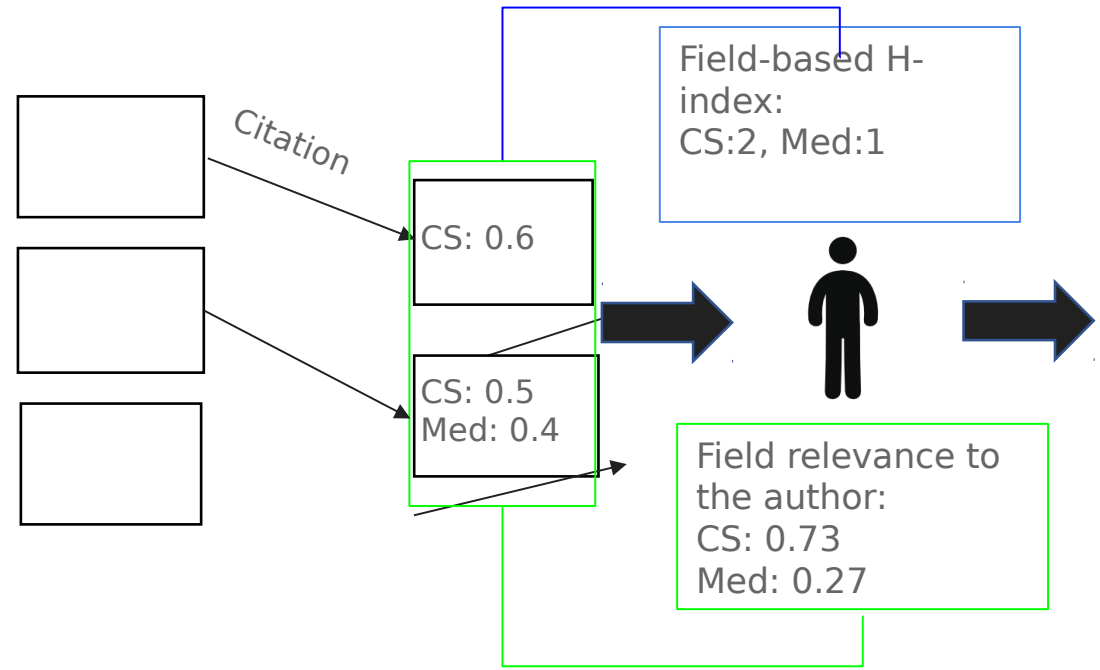
Link to App: http://graphdegeneracy.org/scientometrics/

And it went on like this..

Impact

- Success = Impact and Influence
- Impact is *relative* to the *field*

Use Microsoft Academic Graph (over 200m papers)

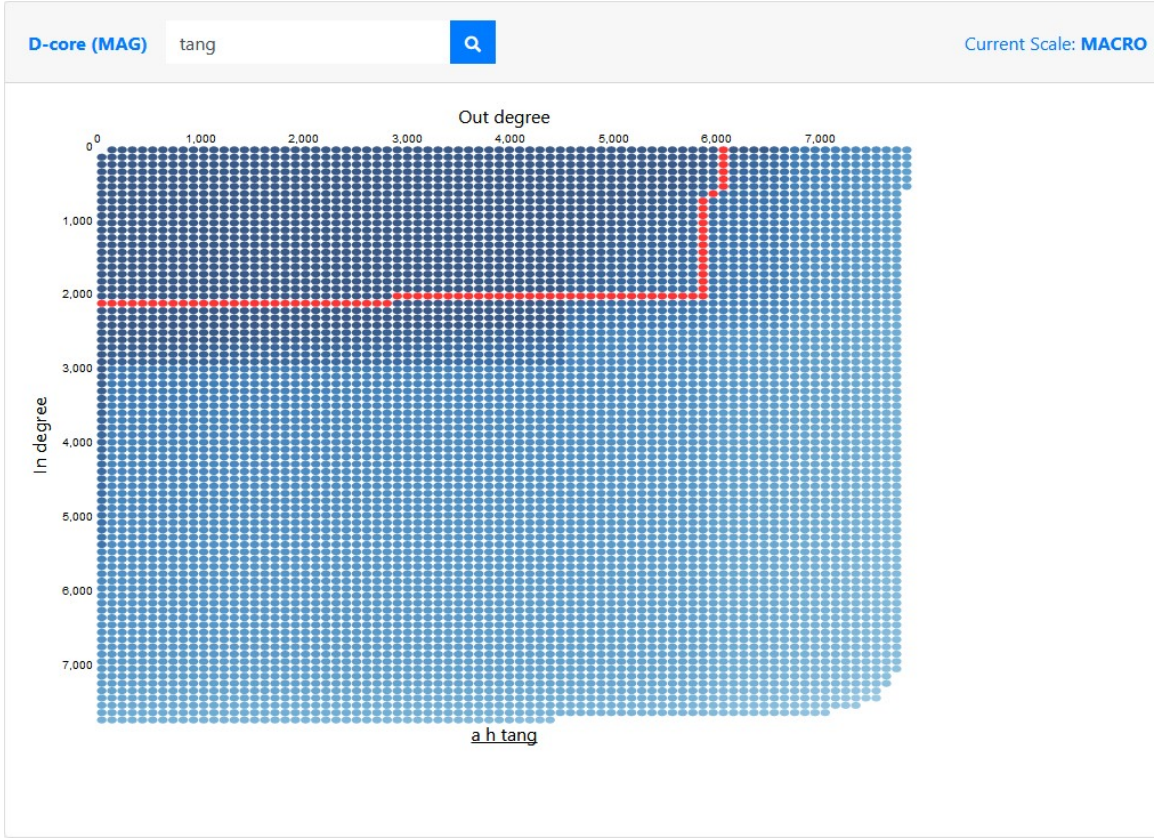




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Influence

- K-core in coauthorship?
  - Wrong, I can have many coauthors without being influential
- Use author-citation network
- Problem 1: Direction is important
  - Have to use D-core
  - More complex and less interpretable
- Problem 2: 17.000.000.000 edges
  - Construct it using the egonetworks
  - Remove edges with < 10 citations
  - Run D-core in parallel



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Proof of COncept

- Looked at the authors in the densest D-core
  - These are mostly high energy physicists
  - Their laboratories collaborate a lot with each other -> the papers have a massive number of authors
  - Every new finding cites previous papers with similar authors

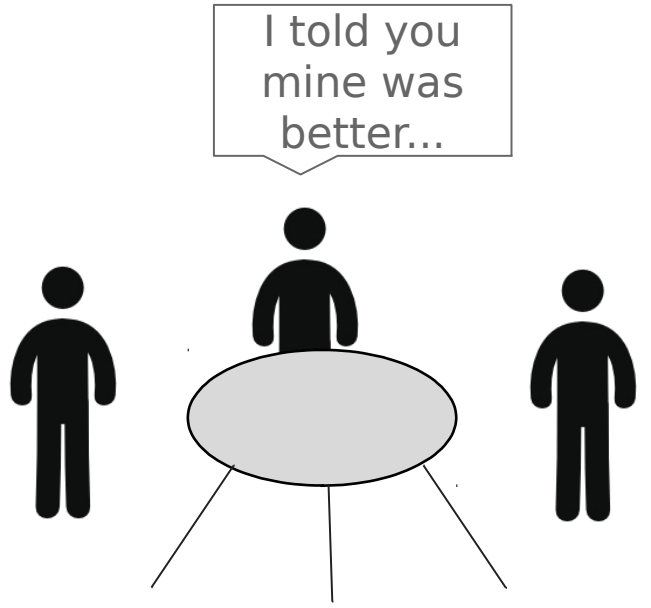


The wordcloud of the paper keywords from the authors in the densest d-core.

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Future work

1. Name disambiguation
2. Self citation rates
3. Number of coauthors



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Thank you

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Code: https://github.com/GiorgosPanagopoulos/MAG-Author-Influence-Scientometrics